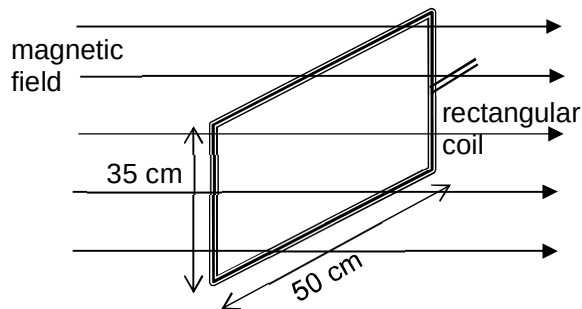


- 6 (a) A simple a.c. generator which consists of a rectangular coil with 40 turns is rotating at 50 revolutions per second in a uniform magnetic field of flux density 50 mT. The coil is 50 cm long and 35 cm wide, as shown in Fig. 6.1.



**Fig. 6.1**

(i) Define *magnetic flux*.

.....  
.....  
..... [1]

(ii) Calculate the maximum e.m.f. induced in the coil.

maximum e.m.f. = ..... V [2]

- (b) The output from the generator in part (a) is connected to the input of a transformer. The transformer has 40 turns on its primary coil and 10 turns on its secondary coil. The transformer may be considered to be ideal. The output from the transformer is connected to a resistor of resistance  $100\ \Omega$  and a diode, as shown in Fig. 6.2.

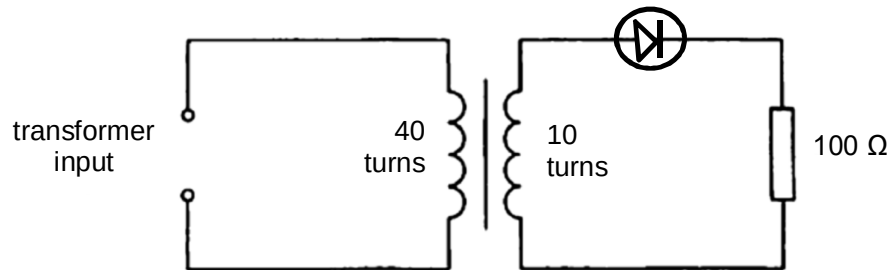
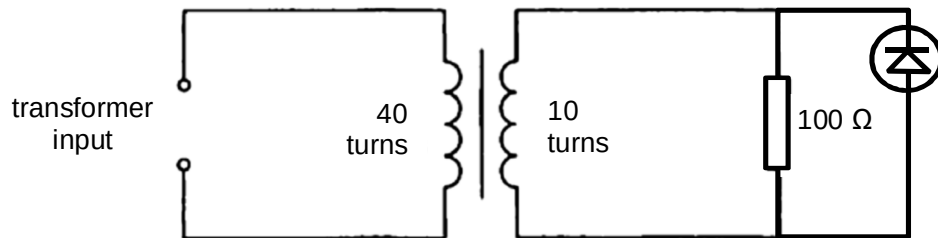


Fig. 6.2

Calculate the root-mean-square voltage across the resistor.

root-mean-square voltage = ..... V [3]

- (c) In an attempt to draw the circuit diagram that represents the system in part (b), a student drew Fig. 6.3 instead.



**Fig. 6.3**

Suggest, with a reason, what will happen if the circuit is set up according to Fig. 6.3.

.....

.....

.....

.....

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.....

.....

[2]